

CSL253 - Theory of Computation

Tutorial 7

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Question 18

A useless state in a pushdown automaton is never entered on any input string. Consider the problem of determining whether a pushdown automaton has any useless states. Formulate this problem as a language and show that it is decidable.

Understanding Useless States

A state in a PDA is said to be **useless** if no input string causes the automaton to enter that state during any computation. Even if the state has transitions leading to or from it, it is considered useless if the transition conditions are never satisfied in real execution.

Proof by Turing Machine Construction (Marking on Tape)

We now describe a Turing machine M that decides whether a pushdown automaton P has any useless states by marking reachable states on its tape.

Input: An encoding of a PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$.

Procedure:

1. On the tape, write the list of all states $q_0, q_1, \dots, q_n$ and mark q_0 as reachable.
2. Repeat:
 - For every marked state q and for every possible input symbol a (including ε) and stack symbol γ (including ε), simulate the transitions $\delta(q, a, \gamma)$.
 - If a transition leads to a new state q' that is unmarked, mark q' .
3. When no new states are marked in a complete pass, halt.

Final Decision:

- If there is at least one unmarked state, M accepts (the PDA has a useless state).
- If all states are marked, M rejects (every state is reachable).

Why this Works:

- The finite set of states guarantees the process halts.
- The machine simulates reachability by nondeterministically exploring the PDA's behavior over stack and input symbols.

Worked Example

Consider a PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, Z, F)$ with:

- $Q = \{q_0, q_1, q_2\}$
- $\Sigma = \{a, b\}$, $\Gamma = \{Z, A\}$
- q_0 is the start state, $F = \{q_1\}$ is the accepting set.

Transitions:

$$\delta(q_0, a, Z) = (q_1, AZ) \quad \text{and} \quad \delta(q_1, b, A) = (q_1, \varepsilon)$$

And an additional dead transition:

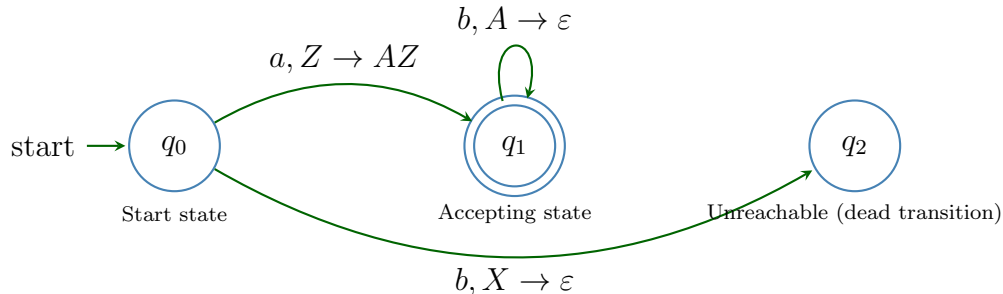
$$\delta(q_0, b, X) = (q_2, \varepsilon)$$

where $X \notin \{Z, A\}$. Since X never appears on the stack, q_2 is never reached.

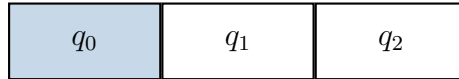
Execution on Turing Machine:

1. Initially mark q_0 .
2. Simulate δ from q_0 . Only q_1 is reached under the transition $a, Z \rightarrow AZ$. Mark q_1 .
3. From q_1 , transitions loop back to q_1 or halt. No new state is reached.
4. Since q_2 is never marked, the Turing machine halts and accepts, proving that q_2 is useless.

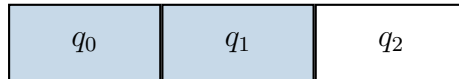
PDA Diagram



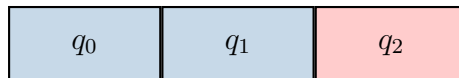
Tape Diagram (Reachability Marking)



Step 0: mark q_0



Step 1: from q_0 , mark q_1



Final: q_2 unreachable